

UNIVERSITÄT
LEIPZIG

The Effect of Visualization on Mental Models of Polarization

Jonas Bley[1], Judith Schmid[1], Fabian Hennig[1], Yonathan White[2], Malte Ubben[1], Avraham Merzel[2], and Philipp Bitzenbauer[1]

[1]Professur für Physikdidaktik
[2]Hebrew University of Jerusalem

Preregistration: osf.io/u6xk9

Overview

1. Mental models and model understanding
2. Context: Polarization and the Poincaré sphere
3. Hypotheses and study design
4. Analysis and current status

Mental models and model understanding

UNIVERSITÄT
LEIPZIG

Gestalt and Functionality

Gestalt

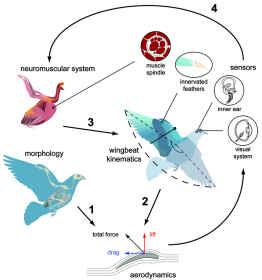
The set of forms and structures that a learner's mental model consists of.

Functionality

The set of functions that a learner's mental model consists of.

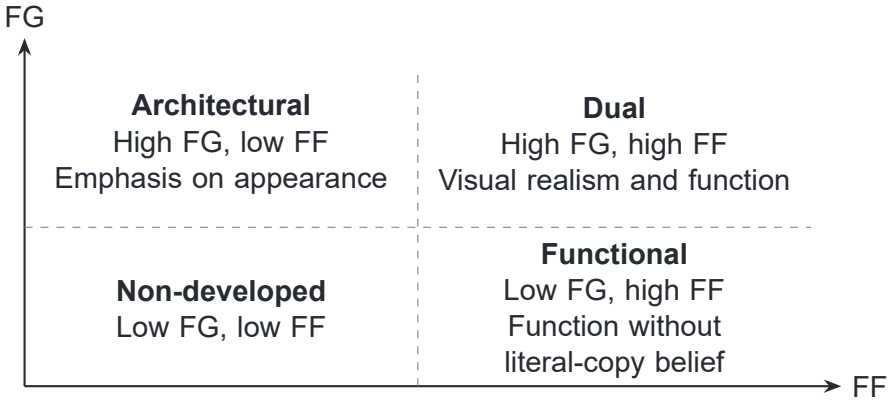


(WikimediaCommons, 2024)



(Altshuler et al., 2015)

Four profiles of fidelity beliefs



(Ubben and Bitzenbauer, 2022; Bitzenbauer and Ubben, 2025)

Fidelity and alignment answer different questions

Fidelity: the learner's stance

- What does the learner believe the model is faithful to?
- FG: appearance; FF: function.

Alignment: expert correspondence

- **Structural/Gestalt alignment (GA):** correspondence with expert-selected forms and structures.
- **Functional alignment (FA):** correspondence with expert-selected functions.

Successful visual reasoning can coexist with literal model beliefs.

Polarization and state-circle study: (Tóth et al., 2024).

Visualization and the multimedia effect

Multimedia principle

Combining words with relevant pictures can benefit learners more than words alone.

(Mayer, 2005)

DeFT: Design, Functions, Tasks

- **Design:** The number of representations, how information is distributed, the form, the order, and how the translations are highlighted.
- **Functions:** Representations can **complement** each other, **constrain** interpretations, and support the construction of **deeper understanding**.
- **Tasks:** The cognitive tasks required for learning with multiple representations: understanding the representations, their relationships, and selecting accordingly.

(Ainsworth, 2006)

Two pathways for an additional visualization

Representational support

- Complement by allowing different strategies to solve problems.
- Connect notation, geometry, and optical transformations.
- Support expert-aligned explanations and predictions.

Spontaneous reflection

- Encounter different representations of the same state.
- Reconsider literal interpretations.
- Attend to shared functions and limits.

Due to the multimedia principle, expert alignment may improve even when fidelities remain stable.

Context: Polarization and the Poincaré sphere

Jones vectors and the Poincaré sphere

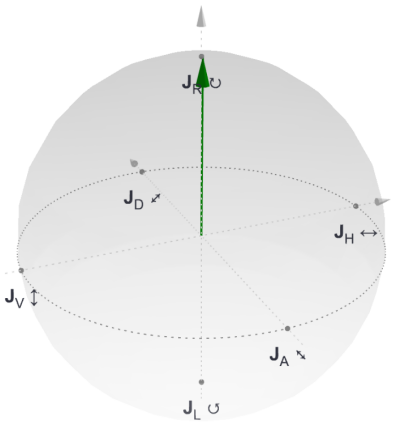
For a normalized, fully polarized state,

$$\mathbf{J} = \begin{pmatrix} a \\ b\,e^{i\varphi} \end{pmatrix}, \qquad a, b \geq 0, \quad a^2 + b^2 = 1.$$

The applet uses

$$\mathbf{S} = \begin{pmatrix} a^2 - b^2 \\ 2ab\cos\varphi \\ 2ab\sin\varphi \end{pmatrix}, \qquad \|\mathbf{S}\| = 1,$$

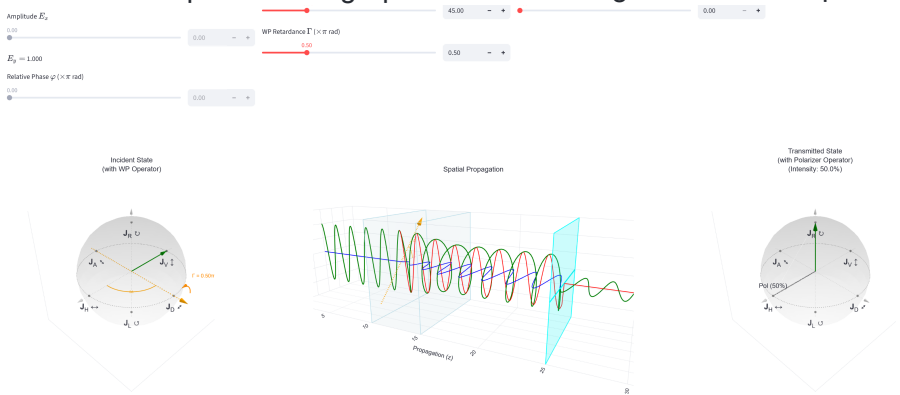
Polarization is a valid approach to introductory quantum computing (Tóth et al., 2025).



The applet

Control:
Jones notation and spatial wave graph.

Experimental:
Adding the Poincaré sphere.



+ Introductory explanations, guided parameter changes, hints, and solutions.

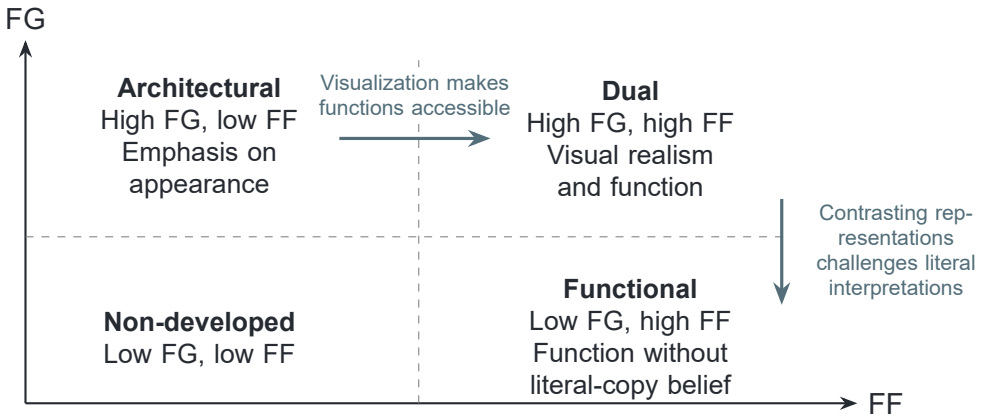
The learning sequence

Steps	Topic	Learner activity
1–3	Polarization states	Change amplitudes and relative phase to construct linear and circular states.
4–6	Wave plates	Vary retardance and fast-axis orientation to transform a state.
7	Linear polarizers	Relate transmission-axis orientation to output intensity.
8–9	Combined operators	Coordinate wave plates and polarizers to meet a target.

Nine challenges precede a completion screen and survey return link.

Hypotheses and study design

Possible changes in fidelity beliefs



Possible pathways, not established transitions (Bitzenbauer and Ubben, 2025).

Research questions

Research Question 1

How does the added Poincaré sphere affect Functional Fidelity (FF) and Fidelity of Gestalt (FG) in learners' models of polarization?

Research Question 2

Does the added sphere improve functional alignment (FA) and structural/Gestalt alignment (GA) with expert views?

Research Question 3

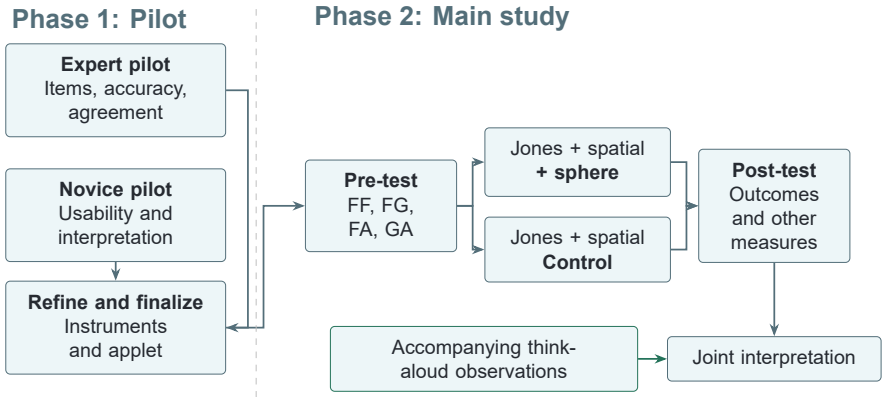
Do effects vary with baseline fidelity beliefs and spatial reasoning? How do reported cognitive load and learning outcomes relate? (Exploratory)

Hypotheses

	Outcome	Sphere versus control, adjusted for baseline
H1a	Functional Fidelity	Higher FF at post-test.
H1b	Fidelity of Gestalt	Lower FG at post-test.
H2	Expert alignment	Higher FA and GA at post-test.

Exploratory: dependence on initial fidelity beliefs and spatial reasoning, associations with cognitive load, and possible learning pathways.

Study structure



Phase 1: Calibrate the instruments and the applet

Expert pilot

- Classify items as FF, FG, both, or neither.
- Judge factual correctness and the usefulness of analogies.
- Evaluate wording and agreement.

Novice pilot

- Identify confusing controls and instructions.
- Examine interpretation of the representations.
- Assess cognitive load (Krieglstein et al., 2023) and technological acceptance (Abdullah and Ward, 2016).

Refine questionnaire items, applet instructions, and interaction design before the main comparison.

Phase 2: Randomized pre/post comparison

Pre-test	FF, FG, FA, and GA
Intervention	Random assignment to guided polarization tasks with or without the added sphere
Post-test	FF, FG, FA, GA; cognitive load (Krieglstein et al., 2023) and spatial reasoning (Young et al., 2025), demographic information

Population: second- and third-year physics / related STEM undergraduates.

Target: Power analysis: $N = 218$ recruited; minimum analyzable sample $N = 196$.
We need your help to recruit participants for the online study!

Measures and protocol decisions

- Cognitive load: Final validated questionnaire with 5 items each for intrinsic, extraneous, and germane load; 9-point scale (Krieglstein et al., 2023)
- General Extended Technology Acceptance Model for E-Learning (GETAMEL, only in pilot study, 7-point Likert scale): (Abdullah and Ward, 2016)
  - Perceived Usefulness
  - Perceived Ease of Use
  - E-Learning Self-Efficacy
  - Perceived Enjoyment
- Spatial reasoning: Measured using the open-source 3D Block Rotation scale from the International Cognitive Ability Resource (ICAR) (Young et al., 2025)

Think-aloud observations: Investigating mechanisms

Observe the students'

- attempts to incorporate the Poincaré sphere into their thinking.
- difficulties in connecting representations.
- Expressed conflicts.
- Conflict resolution.

The observations can provide evidence for mechanisms underlying learning and limitations of mental models.

Analysis and current status

Planned primary analysis: MANCOVA

Joint outcomes

$$\mathbf{Y}_{\text{post}} = \begin{pmatrix} FF_{\text{post}} \\ FG_{\text{post}} \\ FA_{\text{post}} \\ GA_{\text{post}} \end{pmatrix}$$

- **Factor:** condition, with or without the additional Poincaré sphere.
- **Covariates:** baseline FF, FG, FA, and GA.
- **Target:** total condition effect on post-test outcomes (RQ1 and RQ2).

Exploratory (RQ3): moderation by baseline fidelities and spatial reasoning; associations with reported cognitive load.

Student feedback: learning support and usability

Learning support

- Students valued interactive graphics and seeing parameter changes immediately.
- Hints helped, including guidance without revealing the answer.
- Requests included more examples and fuller explanations.

Usability

- Camera resets, perspective controls, and scrolling caused difficulties.
- Requests included typed values and persistent display choices.
- Tasks and plots were hard to view together; earlier formulas were inaccessible.

Summary of open-ended pilot feedback: 8 respondents across questions.

Student feedback: guidance and motivation

Conceptual guidance

- A student solved tasks despite uncertainty about matrices and terminology.
- Requests included a glossary and definitions of polarization and circular handedness.
- One student wanted the purpose and connection to quantum physics explained.

Motivation and interpretation

- Students enjoyed changing the graphics and using the simulation.
- One student compared successive tasks to a game's reward system.

Summary of Open-ended pilot feedback: 8 respondents across questions.

Outlook

Pilot study takeaways:

- Students valued interactivity, graphics, and hints.
- Comments identified usability difficulties and requests for conceptual guidance.
- The comments do not isolate a benefit of the Poincaré sphere.

Next steps:

- Collect the remaining expert data and conduct the main study.
- As physics education experts, we need your help (open until 09th October 2026):



https://nimble.li/omz88r4m

References I

Abdullah, F. and R. Ward (2016). “Developing a General Extended Technology Acceptance Model for E-Learning (GETAMEL) by analysing commonly used external factors”. In: *Computers in Human Behavior* 56, pp. 238–256.

Ainsworth, Shaaron (2006). “DeFT: A conceptual framework for considering learning with multiple representations”. In: *Learning and Instruction* 16.3, pp. 183–198. DOI: 10.1016/j.learninstruc.2006.03.001.

Altshuler, Douglas, Joseph Bahlman, Roslyn Dakin, Andrea Gaede, Benjamin Goller, David Lentink, Paolo Segre, and Dimitri Skandalis (Oct. 2015). “The biophysics of bird flight: functional relationships integrate aerodynamics, morphology, kinematics, muscles, and sensors”. In: *Canadian Journal of Zoology* 93, pp. 961–975. DOI: 10.1139/cjz-2015-0103.

Bitzenbauer, Philipp and Malte S. Ubben (2025). “The structure of learners’ perceptions of models (not only) in quantum physics: spotlight on Fidelity of Gestalt and Functional Fidelity”. In: *EPJ Quantum Technology* 12, p. 16. DOI: 10.1140/epjqt/s40507-025-00316-7.

Bley, Jonas (Aug. 28, 2026). *Polarization of Light Applet*. Version 1.0.0.

Krieglstein, Felix, Maik Beege, Günter Daniel Rey, Christina Sanchez-Stockhammer, and Sascha Schneider (2023). “Development and Validation of a Theory-Based Questionnaire to Measure Different Types of Cognitive Load”. In: *Educational Psychology Review* 35.1, p. 9. DOI: 10.1007/s10648-023-09738-0.

Mayer, Richard E. (2005). “Cognitive Theory of Multimedia Learning”. In: *The Cambridge Handbook of Multimedia Learning*. Ed. by Richard E. Mayer. Cambridge University Press, pp. 31–48. DOI: 10.1017/CBO9780511816819.004.

References II

Rexigel, Eva, Jochen Kuhn, Sebastian Becker, and Sarah Malone (2024). “The More the Better? A Systematic Review and Meta-Analysis of the Benefits of More than Two External Representations in STEM Education”. In: *Educational Psychology Review* 36.4, p. 124. DOI: 10.1007/s10648-024-09958-y.

Tóth, Kristóf, Sergej Faletič, Marisa Michelini, Gesche Pospiech, Andrea Betti, Marco Nicolini, Marco Parmiggiani, Joaquin Veith, and Philipp Bitzenbauer (2025). “Investigating the effect of two-state approaches on students’ understanding of quantum measurement: A quasiexperimental field study”. In: *Physical Review Physics Education Research* 21.2, p. 020142. DOI: 10.1103/15rm-rlnj.

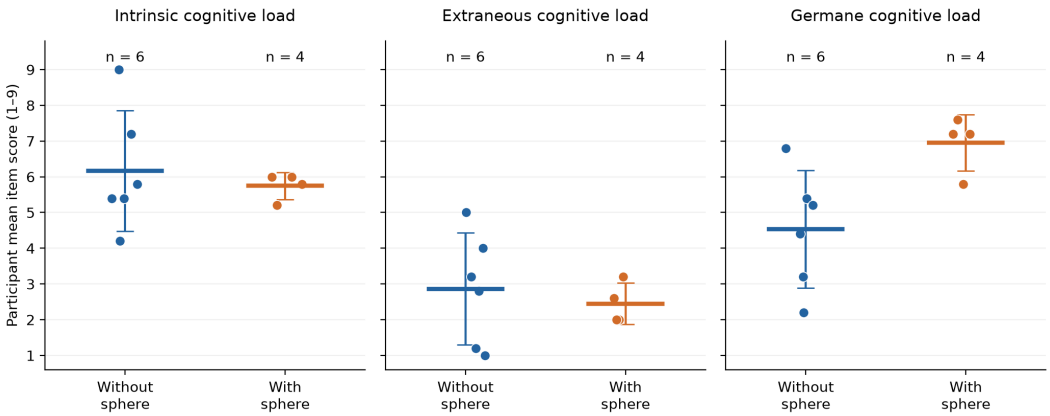
Tóth, Kristóf, Marisa Michelini, and Philipp Bitzenbauer (2024). “From light polarization to quantum physics: Supporting lower secondary school students’ transition from gestalt to functional thinking”. In: *Eurasia Journal of Mathematics, Science and Technology Education* 20.6, em2449. DOI: 10.29333/ejmste/14587.

Ubben, Malte S. and Philipp Bitzenbauer (2022). “Two Cognitive Dimensions of Students’ Mental Models in Science: Fidelity of Gestalt and Functional Fidelity”. In: *Education Sciences* 12.3, p. 163. DOI: 10.3390/educsci12030163.

WikimediaCommons (2024). *File:BIRD.jpg — Wikimedia Commons, the free media repository*. [Online; accessed 17-September-2026].

Young, S. R., J. Kim, K. McKee, D. R. Doyle, M. A. Novack, W. Revelle, R. Gershon, and E. M. Dworak (2025). “Validation of International Cognitive Ability Resource (ICAR) Implemented in Mobile Toolbox (MTB)”. In: *Journal of Intelligence* 13.12, p. 154. DOI: 10.3390/jintelligence13120154.

Initial pilot findings: cognitive load



Initial pilot findings: technological acceptance

